(a)

PINNStudio Examples: Forward | Helmholtz 2D – sound-hard scatte | Default | Open | Save | Export script | Solve | Stop

SETTINGS

BC 3 · Robin, u_re

One boundary condition: its type, the output it applies to, where on the boundary (Where:) and its value.

Condition

Type: Robin BC Output: 1

Where: Custom expression... left or right or bottom or top

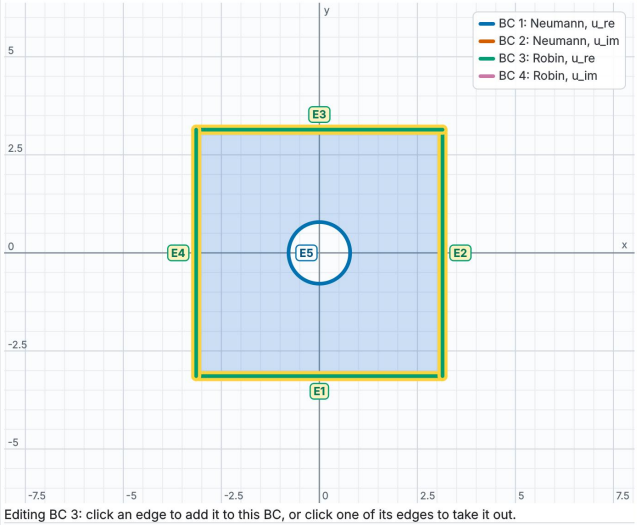
Value: -k0*u_im

On the domain

Picks E1, E2, E3, E4. Click edges in the graphics to add them to this BC or take them out.

Start | Results | Geometry

Fit view



Editing BC 3: click an edge to add it to this BC, or click one of its edges to take it out.

(b)

PINNStudio Examples: Forward | Poisson 3D – cube with spherical | Default | Open | Save | Export script | Solve | Stop

SETTINGS

BC 2 · Neumann, u

One boundary condition: its type, the output it applies to, where on the boundary (Where:) and its value.

Condition

Type: Neumann BC Output: 1

Where: Custom expression... np.isclose(x**2 + y**2 + (z - 1)**2, R**2)

Value: 2*exp(-(x**2 + y**2 + z**2))*(x**2 + y**2 + z*(z - 1))/R

On the domain

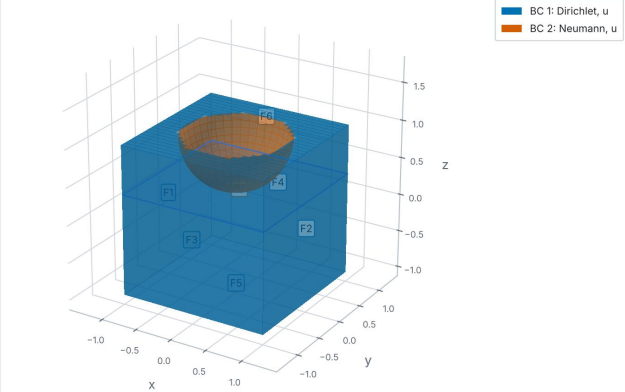
Picks F7. Click faces in the graphics to add them to this BC or take them out.

Start | Results | Geometry

Fit view

View: 3D Slice at z = 0.350

Iso | Top | Front | Side | Show points



Drag to rotate, scroll to zoom. Each face takes the colour of its boundary condition; the translucent plane is the slice the flat views show. Draw shapes and pick faces in a flat view, or pick faces in the list below.